7 Figure 7.1 shows two identical cells in series with a thermistor and a fixed resistor.

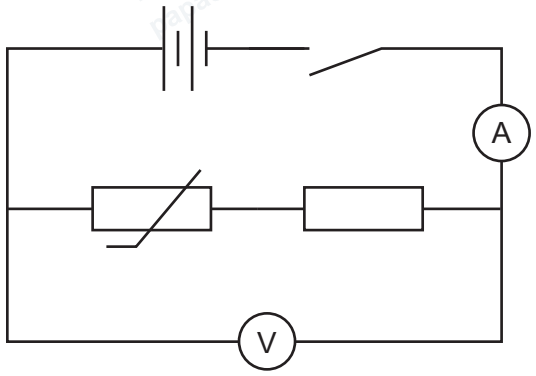

Figure 7.1

The switch is closed. The reading on the voltmeter is 9.0 V. The reading on the ammeter is 4.6 mA.

(a) Determine the e.m.f. of one cell. Give the unit.

e.m.f. = ... unit ..... [2]

(b) Complete the sentence to define electric current.

Electric current is the amount of ..... passing a point

per unit ..... .

[1]

(c) The resistance of the fixed resistor is 480 Ω.

Calculate the resistance of the thermistor.

resistance = ... Ω [3]

(d) The thermistor is placed in a beaker of ice. The temperature of the thermistor decreases. State how this change in temperature affects the readings on the voltmeter and ammeter.

voltmeter .....

ammeter .....

[1]

[Total: 7]

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